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Effects of Mindfulness-Based Stress Reduction Program on Mental Health of Non-Clinical Older Adults: A Systematic Review on Randomized Controlled Trials

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ABSTRACT

Background: Aging may result in many physical and mental health challenges that need to be diagnosed and treated. Untreated mental illness is associated with many major health issues, which may be the cause of premature mortality. Nowadays, many non-pharmacological means are being implemented to find out their benefits for mental and physical wellness. **Objective:** The present study reviewed the randomized controlled trials (RCTs) with the purpose of investigating the effects of the Mindfulness-Based Stress Reduction (MBSR) program on mental health of non-clinical older populations. **Methods:** Initially, 518 studies were identified through five electronic databases such as CINAHL, PubMed, PubMed Central, ScienceDirect, and Scopus among which finally, seven RCTs published between 2009-2018 were considered based on meeting the adopted inclusion criteria. The risk of bias and quality of the studies were assessed by the Jadad scale. **Results:** A total of 997 participants with a mean age of over 40 years from four different countries like the USA, Sweden, Taiwan, and China were included in the current systematic review among them, 482 participated in the MBSR program of 4-8 weeks as intervention group. The studies considered the mental issues viz. anxiety, mindfulness, perceived stress, depression, distress, coping, psychological well-being, quality of life, etc. as dependent variables. The majority of the studies found significant improvement in mindfulness, anxiety, psychological distress, and psychological well-being among non-clinical older adults. **Conclusion:** Despite the variety in the basic features of the MBSR program, this review study concludes that the MBSR program is effective for improving the overall mental health of non-clinical older adults.

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Introduction

The number of older people is increasing rapidly across the world. By the year 2050, the WHO predicts that the percentage of people over 60 will nearly double, rising from 12% to 22% (1). Aging is inevitable and irreversible (2). In consequence of the development of medical science and technology, the life span of human has increased (3) , but at the same time, life has become more sedentary, which results in a variety of health problems and mental illness (4). Apart from that, older people are more likely to deal with loss, a drop in socio-economic status, and physical and economic dependence, which may result in isolation, loneliness, and many psychological problems, especially anxiety, distress, and depression (1). In adults 60 and older, 15% are thought to have a mental illness. Impaired mental wellbeing is associated with an increased risk of physical illnesses such as reduced mobility, chronic pain, frailty, coronary heart disease, diabetes, disability, and premature mortality (5). Mental well-being is a protective factor against unhealthy aging. A study suggests that positive life evaluation, such as happiness, predicts lower future mortality and morbidity (6). So, there is a widely recognized need to prioritise the measures for sustaining mental wellbeing among the elderly. Nowadays, numerous research investigations are being conducted to provide non-pharmacological means to ensure geriatric fitness and wellness. Commonly implemented non-pharmacological means for the elderly include exercise, yoga, mindfulness-based programs etc.

Mindfulness is related to meditation, and it consists of formal meditation exercises as well as informal exercises (7,8). Mindfulness may be defined as an active, nonjudgmental awareness of thoughts, feelings, and sensations as they happen in the present. This practice has been proven to reduce stress, improve cognitive function, and enhance emotional wellbeing (9).

Kabat-Zinn developed the Mindfulness-Based Stress Reduction (MBSR) program which is commonly used as a meditation practice and is implemented by trained professionals. The MBSR program consists of didactics and individual and group practices, which include breathe awareness meditation, body scans, walking meditation, and yoga. Each of the practices involves focusing attention on present-moment experience (10–12). Mindfulness interventions may vary in delivery mode and target population. The duration of intervention may range from long-term (long sessions of 2.5 hours of 10 months) to short term (30-minute sessions of 4 weeks). The intensity of intervention may vary from high dose to low dose (13). As per Kabat-Zinn, the MBSR program consists of eight 2.5-hour weekly sessions and one 7-hour day of silence, followed by 45 minutes of daily home-based sessions (10,11). Training the body and mind through meditation on a daily basis can enhance mood, relax muscles, lower heart rate and blood pressure, relieve tension, reduce lifestyle stress, and activate the release of feel good hormones such as serotonin and dopamine. It also produces gamma-aminobutyric acid (GABA), which helps in decreasing anxiety and stress and lowers the secretion of norepinephrine, thereby reducing anxiety (14,15). Further, meditation practice results in a reduced cortisol level that may lead to lower anxiety and depression (16,17). Many research studies reveal that MBSR is associated with significant declines in anxiety, depression, and distress, as well as significant increases in mindfulness, psychological well-being, and quality of life (18–22).

The purpose of the current review was to investigate the randomized controlled trials on the effects of mindfulness-based stress reduction program on the mental health of non-clinical older populations.

Materials & Methodology

Search Strategy

For the current review, the electronic databases CINAHL, PubMed, PubMed Central, ScienceDirect, and Scopus were thoroughly searched from November 2022 to February 2023. The articles were searched from 2009 to 2022. The searching terms were "mindfulness meditation and mental health", "mindfulness and older adults", "older adults and mental health," and "mindfulness or mental health or older adults." The electronic database search yielded 518 articles, among which 21 were initially found eligible for final screening. Finally, seven articles were included in this review for analysis. The systematic search of the present study was conducted according to Preferred Reporting Systematic Reviews (PRISMA) (23) without meta-analysis and depicted in the PRISMA flow chart (Figure 1).

Inclusion Criteria:

The relevant studies were considered which investigated the effect of a mindfulness-based stress reduction (MBSR) program on the mental health of non-clinical older adults. The articles were finally included in the present review which met the following inclusion criteria:

(a) Articles: All the articles published in English from 2009 to 2022 through five selected databases (CINAHL, PubMed, PubMed Central, ScienceDirect, and Scopus).

(b) Participants: The non-clinical older adults of mean age 40 years and above.

(c) Intervention: Only the MBSR program.

(d) Outcomes: The investigations that measured the mental health related psychological components, like anxiety, distress, depression, mindfulness, mental well-being, quality of life, etc.

(e) Study design: Only randomized controlled trials (RCT).

Exclusion Criteria:

Older adult patients suffer from chronic physical illnesses as well as major psychological disorders.

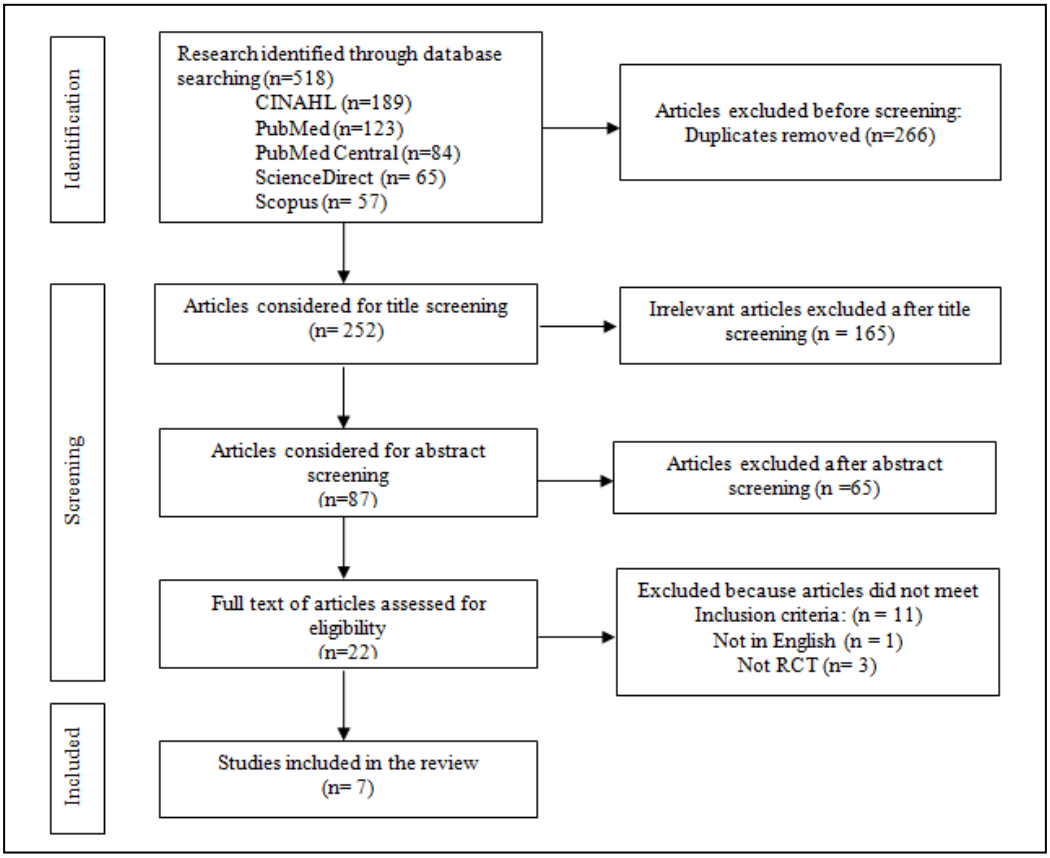


Figure 1. Search strategy and study selection process (PRISMA 2020 flow diagram)

Risk of bias and Quality Assessment:

The Jadad scale was used for assessing the risk of bias and quality of the studies in the present review (24). The Jadad scale defines the methodological features of the RCTs, which consists of seven questions. Each question was answered with a "yes" or "no" option. Every "yes" answer for the first five questions added one point, while one point was subtracted for a "yes" answer to the remaining two questions. The Jadad score ranges from 0-5, in which a 0–2 score indicates "low" quality and a 3-5 score indicates "high" "quality. The studies had been assessed by the three reviewers, and if there was any conflict among the reviewers, it was solved by discussion.

Data extraction:

The data extraction of the included studies in the current review was done by the two reviewers (TD and AB) individually. The categories of the data were coded as follows: authors, year, country, participants (sample size, groups, gender), mean age, intervention descriptions, outcome measures related to psychological health, and quality of the study (Jadad score). The final recommendation of the changes in coded variables was done after consulting with the first reviewer (NB).

Results

Summary of included studies:

A total number of 518 studies were searched through five databases, among which only seven (n=7) studies (18,21,25–29) met the inclusion criteria and were included in the present review. All studies were RCTs that had been published from 2009 to 2018. A total of 997 participants from four countries, viz., the USA, Sweden, Taiwan, and China, participated; amongst them, 82.71% were women. Out of 997 participants, only 482 participated in the

Table 1. Characteristics of included studies.

Authors, Year, Country	Participants (Sample size, Group, Gender)	Mean age (SD)	MBSR intervention (duration, no of session, session length)	Dependent variables	Outcomes	Jadad Score (range 0-5)
Carmody et al., 2011, (18) SA	s= 110, MBSR group= 57, WLC= 53, Women (100%)	MBSR group= 52.5 (5.4) WLC= 53.8 (4.4)	8 weeks, Weekly 2.5 hours class, 45 min home practice 6 days/week.	Quality of life, anxiety, perceived stress	Significant improvement in quality of life (p=0.022), anxiety (p=0.005), and perceived stress (p= 0.001) among MBSR group.	3
Huang et al., 2015, (25) Taiwan	s=144, Intervention = 72, Control= 72, Women (41%)	Intervention = 42.4y, Control= 42.7y.	8 weeks, 2 hours weekly training, 45 min daily home practice.	Psychological distress, perceived stress	Significantly lower psychological distress (p=.000), perceived stress (p=.005) in intervention group compared to control group after completion of MBSR intervention.	3
Josefsson et al., 2012, (21) Sweden	s= 126, Mindfulness (n= 46), Relaxation (n= 40), Wait-list (n= 40), Women (91%)	Mindfulness = 48.9y (10.2), Relaxation = 50.4y (10.4), Wait-list = 45.1y (8.9).	4 weeks, 2sessions/week, 45 min session.	Anxiety, depression, coping, mindfulness, psychological well-being	No significant differences in anxiety, depression and coping. Significant changes in FFMQ total ($p<0.01$) and PWB total ($p<0.01$) scores among mindfulness and relaxation group compared to wait-list group.	3
Malarkey et al., 2013, (26) USA	s= 186, MBI-Id (n= 93), Education control (n=93), Women (88%)	MBI-Id = 51y, Education control = 49y	8 weeks, Group meeting 1hour/week, 20 min daily practice.	Perceived stress, depression, mindfulness	No significant changes in perceived stress and depression at 2-months but significant changes in mindfulness among MBI-Id group compared to education control and the changes were sustained at 6 and 12 months (p=0.03).	3

Moynihan et al., 2013, (27) USA	s=201, MBSR group= 100 WLC= 101, Women (62%)	Mindfulness group= 73.3y (6.7) Wait-list control= 73.6y (6.7)	8 weeks, Weekly 120 min sessions and one 'all day intensive' session of 7 hour.	Perceived stress, depression, mindfulness	Significant improvement in mindfulness (p=0.023) and no improvement in perceived stress or depressive symptoms among MBSR group.	3
Pipe et al., 2009, (28) USA	s= 33, Mindfulness meditation course= 16, Control= 17, Women (97%)	Mindfulness meditation course= 50.2 (6.56), Control= 49.4 (7.11)	4 weeks, 5 sessions of 2 hours, 30 min daily practice.	Distress, anxiety	Significant improvement was found in Positive symptom distress index (p=0.010), anxiety (p= 0.0409) among mindfulness meditation group.	3
Wong et al., 2018, (29) China	s= 197, MBSR group= 98, Menopause education control= 99, Women (100%)	MBSR group= 51.9 (3.0), Menopause education control= 52.1 (3.2)	8 weeks, Weekly 2.5 hours sessions, 40 min home practice daily.	Anxiety, Depression, perceived stress, mindfulness	Significant changes in anxiety (p=0.007), depression (p=0.031), and mindfulness-observe (p=0.016) but no significant changes in perceived stress (p>0.05) at 8 months among MBSR group (dependent variables were assessed after 6 months of 8 weeks of intervention).	3

Note: SD= Standard deviation, s= participants, y= years, min= minutes, MBSR=Mindfulness-Based Stress Reduction, WLC= wait-list control, MBI-Id= Mindfulness-Based Intervention, FFMQ= Five-Facet Mindfulness Questionnaire, PWB= Psychological Well-being, p= level of significance.

MBSR program and the remaining 515 participants served as controls and in other groups. The mean age of the participants ranged between 42.4-73.6 years. All of the included studies selected the MBSR program (duration ranged from 4–8 weeks) as an independent variable to find out the effects on the mental health of older adults, where various psychological health components were assessed like quality of life, anxiety, perceived stress, depression, distress, coping, mindfulness, psychological well-being, etc. The characteristics of the included studies are depicted in Table 1.

Risk of bias and Quality Assessment:

The Jadad score results are depicted in Table 1. The result shows that all of the included studies scored 3 points, which indicates each of those studies was of "high" quality. Though the present review considered the articles that were published only in English and the results of the quality of all studies were rated high, the probability of publication bias is likely to be present there.

Discussion

The aim of the current review was to investigate the experimental randomized controlled trials on the effect of the MBSR program on non-clinical older adults. With this aim, after searching through five databases, only seven RCTs published in English from four countries were considered. All 7 studies used the MBSR program as an independent variable for 4–8 weeks along with daily home-based practice of 20–45 minutes. Only one study used one 7-hour day of silence session (27) along with weekly sessions and daily home-based practice. Out of seven studies, only two (21,28) used the 4-week MBSR program, and the remaining studies used an 8-week program. Studies considered anxiety, mindfulness, perceived stress, depression, distress, coping, psychological well-being, quality of life, etc. as

dependent variables. Out of 7 studies 57% study (18,21,28,29) observed the effect of MBSR program on anxiety and among them 75% study (18,28,29) found significant decrease after completing the program. In case of mindfulness only 4 studies (21,26,27,29) observed the effect and all found significant improvement following MBSR program, whereas 5 studies (18,25–27,29) used perceived stress as dependent variable but only 2 studies (18,25) obtained significant reduction among the participants. The other dependent variable such as distress was observed by 2 studies and found significant reduction. One study (18) used quality of life and one (21) psychological well-being as dependent variable and both studies found significant improvement in the said parameters. But four (21,26,27,29) out of seven studies incorporated depression as parameter and only 1 study (29) reported significant improvement, where as only 1 study (21) included coping as psychological parameter and found no significant changes. Three (21,25,26) out of seven studies found significant improvement in mindfulness, PWB, psychological distress, and perceived stress among the participants of the intervention group in comparison to their control participants. In consistent with our findings, an experimental study implemented the MBSR program of 8 weeks on middle-aged women and found significant improvement in PWD and distress (22), and another study found the effect of the MBSR program on older adults and found significant improvement in perceived stress and PWB. In a recent systematic review, Galante et al. reported that mindfulness-based programs were effective in improving average anxiety, depression, and distress compared with no intervention group (30).

We have found mixed results in current review due to fewer empirical studies, inconsistency in applied MBSR intervention (length of sessions: 1 hour to 2.5 hours; number of weekly sessions: one to two; duration of program: 4 to 8 weeks; daily practice- 20 to 45 minutes; a few studies also omitted all-day retreat), heterogeneous population (two out of seven studies employed only menopausal and post-menopausal women as their study participants), etc. Keeping consistency with our findings, Geiger et al. (31) found the same mixed results based on their review of the effects of mindfulness programs on older adults. So, it may be suggested for more future experimental research to be conducted on the MBSR program to establish its effect more strongly on mental health of a healthy population.

Study Limitation:

The review was conducted only on the MBSR program where many other non-pharmacological interventions may benefit the mental health. Only healthy older populations above the mean age of 40 years were considered, while the MBSR program may contribute benefits to patients suffering from mental disorders. The studies published in English were only included, and thereby the study might have publication bias. Further, the study population was not homogeneous in nature. Moreover, only a very few experimental studies were found on the MBSR program for healthy elderly people.

Conclusion

The present review included only RCTs. The most commonly used meditation program, the MBSR was considered as intervention program and the subjective tools used for measuring dependent variables were standard and globally accepted. Moreover, all high-quality studies were included. Though there was a wide range of heterogeneity in the basic features of the MBSR intervention protocol and dependent variables, the current review may conclude that the MBSR program is effective for improving the overall mental health of older adults. Moreover it may be used in community settings as well as home-based settings and in the medical and rehabilitation sectors as an alternative complementary medicine. However, further research needs to be conducted so that the elderly population may use the program as a preventive measure rather than as rehabilitative treatment.

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